

Section: Module 3

Average Treatment Effect

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- Please check the solution of Problem Set 2
 - Great job everyone
 - I know conditional randomization test is hard, so please take a look at the solution!
 - It is important to understand the idea
- Today's agenda
 - Review of mathematical tools: expectation and variance
 - Two inference frameworks: finite-population and super-population
 - Estimand and Estimator
 - Difference-in-Means Estimator
 - Neyman Variance Estimator
 - Efficiency gains from blocking

Mathematical Tool: Variance

- Variance for random variable X_i is defined as

$$\mathbb{V}[X_i] = \mathbb{E}[(X_i - \mathbb{E}[X_i])^2]$$

- Another useful representation

$$\mathbb{V}[X_i] = \mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2$$

- Properties

- $\mathbb{V}[X_i + Y_i] = \mathbb{V}[X_i] + \mathbb{V}[Y_i] + 2\text{Cov}(X_i, Y_i)$
- If a is constant, $\mathbb{V}[a] = 0$
- If a is constant, $\mathbb{V}[aX_i] = a^2\mathbb{V}[X_i]$
- **Law of Total Variance:**

$$\mathbb{V}[X_i] = \mathbb{V}[\mathbb{E}[X_i \mid Y_i]] + \mathbb{E}[\mathbb{V}[X_i \mid Y_i]]$$

Two inference frameworks: Overview

- **within sample inference** (finite-population framework)

- Given data on n units ($i = 1, \dots, n$), we are interested in the average treatment effect on that sample
- Only source of randomness is *treatment assignment*
- Estimand: Sample Average Treatment Effect (SATE)

$$\tau_{SATE} = \frac{1}{N} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}$$

- **population inference** (super-population framework)

- Generalizing inference on the sample to some population of interest
- Source of randomness is (1) *treatment assignment* and (2) *sampling*
- Estimand: Population Average Treatment Effect (PATE)

$$\tau_{PATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Two inference frameworks: Overview

Population
(e.g., all students
in class)



Sampling

Finite
Population
(e.g., students in
section)

Y(1)	Y(0)
10	5
3	7
2	9



Treatment
Assignment

Observed
Data

T	Y
1	10
0	7
1	2

Estimand and Estimator

- Can you differentiate τ , $\hat{\tau}$, $\mathbb{V}[\hat{\tau}]$, and $\widehat{\mathbb{V}}[\hat{\tau}]$?

- τ : Estimand for Average Treatment Effect

- **Estimand**: Target quantity of interest

$$\tau_{SATE} = \frac{1}{N} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}, \quad \tau_{PATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$$

- $\hat{\tau}$: Estimator of Average Treatment effect

- **Estimator**: something you can calculate from data

$$\hat{\tau}_{SATE} = \hat{\tau}_{PATE} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i$$

- $\mathbb{V}[\hat{\tau}]$: **Estimand** of Variance of ATE Estimator

- $\widehat{\mathbb{V}}[\hat{\tau}]$: **Estimator** of Variance of ATE Estimator

- Take a squared root to get standard error

Difference-in-Means Estimator

- Recall that estimand for PATE and SATE is

$$\tau_{SATE} = \frac{1}{N} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}, \quad \tau_{PATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$$

- Difference-in-Means estimator is

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i$$

which is **unbiased** for both PATE and SATE

- Justification for point estimate

Unbiasedness Proof (SATE) - (1)

- For SATE, we need to show $\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{SATE}$.
- By consistency,

$$\begin{aligned}\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] &= \mathbb{E}\left[\frac{1}{n_1} \sum_{i=1}^n T_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i \mid \mathcal{O}_n\right] \\ &= \mathbb{E}\left[\frac{1}{n_1} \sum_{i=1}^n T_i Y_i(1) - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i(0) \mid \mathcal{O}_n\right]\end{aligned}$$

- Because we condition on $\mathcal{O}_n = \{Y_i(1), Y_i(0)\}_{i=1}^n$, potential outcomes are not random.

$$= \frac{1}{n_1} \sum_{i=1}^n \mathbb{E}[T_i \mid \mathcal{O}_n] Y_i(1) - \frac{1}{n_0} \sum_{i=1}^n \mathbb{E}[1 - T_i \mid \mathcal{O}_n] Y_i(0)$$

Unbiasedness Proof (SATE) - (2)

- By complete randomization, we have $Y_i(1), Y_i(0) \perp T_i$. So,

$$= \frac{1}{n_1} \sum_{i=1}^n \mathbb{E}[T_i] Y_i(1) - \frac{1}{n_0} \sum_{i=1}^n \mathbb{E}[1 - T_i] Y_i(0)$$

- Now, $\mathbb{E}[T_i] = \frac{n_1}{n}$ and $\mathbb{E}[1 - T_i] = \frac{n_0}{n}$. So,

$$\begin{aligned} &= \frac{1}{n_1} \frac{n_1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n_0} \frac{n_1}{n} \sum_{i=1}^n Y_i(0) \\ &= \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\} = \tau_{\text{SATE}} \end{aligned}$$

Unbiasedness Proof (PATE)

- We have shown $\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{SATE}$
- We next show $\mathbb{E}[\hat{\tau}] = \tau_{PATE} = \mathbb{E}[Y_i(1) - Y_i(0)]$
- Recall law of iterated expectation: $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$. Thus,

$$\mathbb{E}[\hat{\tau}] = \mathbb{E}[\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n]] = \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}\right]$$

- By linearity of expectation (i.e., $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$),

$$= \frac{1}{N} \sum_{i=1}^n \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1) - Y_i(0)] = \tau_{PATE}$$

Tips for Proof

- Think about which variable is random
 - e.g., In finite-population framework, $Y_i(t)$ is fixed and T_i is random
- Think about the setting
 - e.g., which variables are independent of other variables?
- Construct your proof step by step
 - Clearly show how you did each transformation in each line
 - Do not use multiple operations in one line
 - e.g., use consistency $\rightarrow$ use randomization $\rightarrow \dots$
- Always law of iterated expectation is your friend

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$$

Neyman Variance Estimator

- We use difference-in-means estimator for both PATE and SATE
- We want to know the variance of the difference-in-means estimator
 - $\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n]$ for SATE and $\mathbb{V}[\hat{\tau}]$ for PATE
- From pre-recorded lecture, we know that

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right)$$

$$\mathbb{V}[\hat{\tau}] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}$$

$$\text{where } S_t^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i(t) - \bar{Y}(t))^2$$

$$S_{01} = \frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \bar{Y}(1))(Y_i(0) - \bar{Y}(0))$$

- Please try by review question 3 by yourself first!

Variance Proof (SATE) - Question 1

- We begin by “centering” the treatment variable. Let

$$D_i = \frac{nT_i}{n_1} - 1.$$

- Under complete randomization, show

$$\mathbb{E}[D_i \mid \mathcal{O}_n] = 0,$$

$$\mathbb{E}[D_i^2 \mid \mathcal{O}_n] = \frac{n_0}{n_1},$$

$$\mathbb{E}[D_i D_j \mid \mathcal{O}_n] = -\frac{n_0}{n_1(n-1)} \quad \text{for any } i \neq j.$$

Variance Proof (SATE) - Answer 1 (1)

- Under complete randomization, n_1 and n_0 are fixed, with $n = n_1 + n_0$. Conditional on $\mathcal{O}_n$, only treatment assignment is random:

$$\mathbb{E}[T_i \mid \mathcal{O}_n] = \mathbb{P}(T_i = 1 \mid \mathcal{O}_n) = \frac{n_1}{n}.$$

- Use $D_i = nT_i/n_1 - 1$ and linearity of expectation:

$$\begin{aligned}\mathbb{E}[D_i \mid \mathcal{O}_n] &= \frac{n}{n_1} \mathbb{E}[T_i \mid \mathcal{O}_n] - 1 \\ &= \frac{n}{n_1} \cdot \frac{n_1}{n} - 1 \\ &= 0.\end{aligned}$$

Variance Proof (SATE) - Answer 1 (2)

- The centered indicator takes two values:

$$D_i = \begin{cases} n_0/n_1 & \text{if } T_i = 1, \\ -1 & \text{if } T_i = 0. \end{cases}$$

- Apply the law of total expectation:

$$\begin{aligned} \mathbb{E}[D_i^2 \mid \mathcal{O}_n] &= \mathbb{E}[D_i^2 \mid T_i = 1, \mathcal{O}_n] \mathbb{P}(T_i = 1 \mid \mathcal{O}_n) \\ &\quad + \mathbb{E}[D_i^2 \mid T_i = 0, \mathcal{O}_n] \mathbb{P}(T_i = 0 \mid \mathcal{O}_n) \\ &= \left(\frac{n_0}{n_1}\right)^2 \frac{n_1}{n} + (-1)^2 \frac{n_0}{n} \\ &= \frac{n_0}{n} \left(\frac{n_0}{n_1} + 1\right) \\ &= \frac{n_0}{n_1}. \end{aligned}$$

Variance Proof (SATE) - Answer 1 (3)

- Assignments are **dependent**: treating one unit leaves $n_1 - 1$ treatment slots among the other $n - 1$ units.
- For $i \neq j$, calculate the joint treatment probability:

$$\begin{aligned}\mathbb{E}[T_i T_j \mid \mathcal{O}_n] &= \mathbb{P}(T_i = T_j = 1 \mid \mathcal{O}_n) \\ &= \mathbb{P}(T_i = 1 \mid T_j = 1, \mathcal{O}_n) \mathbb{P}(T_j = 1 \mid \mathcal{O}_n) \\ &= \frac{n_1 - 1}{n - 1} \cdot \frac{n_1}{n} \\ &= \frac{n_1(n_1 - 1)}{n(n - 1)}.\end{aligned}$$

Variance Proof (SATE) - Answer 1 (4)

- Substitute $D_i = nT_i/n_1 - 1$ and expand the product:

$$\begin{aligned}\mathbb{E}[D_i D_j \mid \mathcal{O}_n] &= \mathbb{E} \left[\left(\frac{n}{n_1} T_i - 1 \right) \left(\frac{n}{n_1} T_j - 1 \right) \mid \mathcal{O}_n \right] \\ &= \frac{n^2}{n_1^2} \mathbb{E}[T_i T_j \mid \mathcal{O}_n] - \frac{n}{n_1} \mathbb{E}[T_i + T_j \mid \mathcal{O}_n] + 1.\end{aligned}$$

- Substitute the joint probability and the two marginal means:

$$\mathbb{E}[D_i D_j \mid \mathcal{O}_n] = \frac{n^2}{n_1^2} \cdot \frac{n_1(n_1 - 1)}{n(n - 1)} - \frac{n}{n_1} \cdot \frac{2n_1}{n} + 1.$$

Variance Proof (SATE) - Answer 1 (5)

- Simplify one step at a time, using $n - n_1 = n_0$:

$$\begin{aligned}\mathbb{E}[D_i D_j \mid \mathcal{O}_n] &= \frac{n^2}{n_1^2} \cdot \frac{n_1(n_1 - 1)}{n(n - 1)} - \frac{n}{n_1} \cdot \frac{2n_1}{n} + 1 \\&= \frac{n(n_1 - 1)}{n_1(n - 1)} - 1 \\&= \frac{nn_1 - n - n_1n + n_1}{n_1(n - 1)} \\&= -\frac{n - n_1}{n_1(n - 1)} \\&= -\frac{n_0}{n_1(n - 1)}.\end{aligned}$$

Variance Proof (SATE) - Question 2

- Define

$$X_i = Y_i(1) + \frac{n_1}{n_0} Y_i(0).$$

- Show that this simplifies the variance expression as follows:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n D_i X_i \right)^2 \mid \mathcal{O}_n \right].$$

Variance Proof (SATE) - Answer 2 (1)

- The difference-in-means estimator is **unbiased**:

$$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{SATE}, \quad \tau_{SATE} = \frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}.$$

- Center the estimator at its conditional expectation:

$$\begin{aligned} \mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] &= \mathbb{E} \left[(\hat{\tau} - \mathbb{E}[\hat{\tau} \mid \mathcal{O}_n])^2 \mid \mathcal{O}_n \right] \\ &= \mathbb{E} \left[(\hat{\tau} - \tau_{SATE})^2 \mid \mathcal{O}_n \right]. \end{aligned}$$

- It remains to show

$$\hat{\tau} - \tau_{SATE} = \frac{1}{n} \sum_{i=1}^n D_i X_i, \quad X_i = Y_i(1) - \frac{n_1}{n_0} Y_i(0).$$

Variance Proof (SATE) - Answer 2 (2)

- Write out the estimator and subtract SATE:

$$\begin{aligned}\hat{\tau} - \tau_{SATE} &= \left\{ \frac{1}{n_1} \sum_{i=1}^n T_i Y_i(1) - \frac{1}{n_0} \sum_{i=1}^n (1 - T_i) Y_i(0) \right\} \\ &\quad - \left\{ \frac{1}{n} \sum_{i=1}^n Y_i(1) - \frac{1}{n} \sum_{i=1}^n Y_i(0) \right\}.\end{aligned}$$

- Combine the four sums into one:

$$\begin{aligned}\hat{\tau} - \tau_{SATE} &= \sum_{i=1}^n \left\{ \frac{1}{n_1} T_i Y_i(1) - \frac{1}{n_0} (1 - T_i) Y_i(0) - \frac{1}{n} Y_i(1) + \frac{1}{n} Y_i(0) \right\} \\ &= \frac{1}{n} \sum_{i=1}^n \left\{ \frac{n}{n_1} T_i Y_i(1) - \frac{n}{n_0} (1 - T_i) Y_i(0) - Y_i(1) + Y_i(0) \right\}.\end{aligned}$$

Variance Proof (SATE) - Answer 2 (3)

- Expand $-(1 - T_i)$ and collect the terms in $Y_i(0)$:

$$\begin{aligned}\hat{\tau} - \tau_{SATE} &= \frac{1}{n} \sum_{i=1}^n \left\{ \frac{n}{n_1} T_i Y_i(1) + \frac{n}{n_0} T_i Y_i(0) - Y_i(1) + \left(1 - \frac{n}{n_0}\right) Y_i(0) \right\}.\end{aligned}$$

- Since $n = n_1 + n_0$, $1 - n/n_0 = -n_1/n_0$. Thus,

$$\begin{aligned}\hat{\tau} - \tau_{SATE} &= \frac{1}{n} \sum_{i=1}^n \left\{ \frac{n}{n_1} T_i Y_i(1) + \frac{n}{n_0} T_i Y_i(0) - Y_i(1) - \frac{n_1}{n_0} Y_i(0) \right\} \\ &= \frac{1}{n} \sum_{i=1}^n \left(Y_i(1) + \frac{n_1}{n_0} Y_i(0) \right) \left(\frac{n}{n_1} T_i - 1 \right) = \frac{1}{n} \sum_{i=1}^n D_i X_i.\end{aligned}$$

Variance Proof (SATE) - Answer 2 (4)

- Substitute the estimation error into the variance:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \mathbb{E} \left[\left(\frac{1}{n} \sum_{i=1}^n D_i X_i \right)^2 \mid \mathcal{O}_n \right].$$

- Move the fixed factor $1/n^2$ outside the expectation:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n D_i X_i \right)^2 \mid \mathcal{O}_n \right].$$

Variance Proof (SATE) - Question 3

- Use the above results to show

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{n_0}{n(n-1)n_1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

- **Hint:**

$$\left(\sum_{i=1}^n X_i \right)^2 = \sum_{i=1}^n X_i^2 + \sum_{i \neq j} X_i X_j.$$

Variance Proof (SATE) - Answer 3 (1)

- Conditional on $\mathcal{O}_n$, X_i is fixed. Expand the square using the hint:

$$\begin{aligned}\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] &= \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n D_i X_i \right)^2 \mid \mathcal{O}_n \right] \\ &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n D_i^2 X_i^2 + \sum_{i \neq j} (D_i X_i)(D_j X_j) \mid \mathcal{O}_n \right].\end{aligned}$$

- Take the fixed quantities outside the expectations:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n^2} \left\{ \sum_{i=1}^n \mathbb{E}[D_i^2 \mid \mathcal{O}_n] X_i^2 + \sum_{i \neq j} \mathbb{E}[D_i D_j \mid \mathcal{O}_n] X_i X_j \right\}.$$

Variance Proof (SATE) - Answer 3 (2)

- Substitute the two moments from Question 1:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n^2} \left\{ \frac{n_0}{n_1} \sum_{i=1}^n X_i^2 - \frac{n_0}{n_1(n-1)} \sum_{i \neq j} X_i X_j \right\}.$$

- Factor out $n_0/\{n(n-1)n_1\}$:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{n_0}{n(n-1)n_1} \left\{ \frac{n-1}{n} \sum_{i=1}^n X_i^2 - \frac{1}{n} \sum_{i \neq j} X_i X_j \right\}.$$

- Rearrange the hint to eliminate the cross-products:

$$\sum_{i \neq j} X_i X_j = \left(\sum_{i=1}^n X_i \right)^2 - \sum_{i=1}^n X_i^2.$$

Variance Proof (SATE) - Answer 3 (3)

- Substitute the cross-product identity:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{n_0}{n(n-1)n_1} \left[\frac{n-1}{n} \sum_{i=1}^n X_i^2 - \frac{1}{n} \left\{ \left(\sum_{i=1}^n X_i \right)^2 - \sum_{i=1}^n X_i^2 \right\} \right].$$

- Combine the two coefficients on $\sum_i X_i^2$:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{n_0}{n(n-1)n_1} \left\{ \sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \right\}.$$

Variance Proof (SATE) - Answer 3 (4)

- Since $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, we have $\sum_{i=1}^n X_i = n\bar{X}$. Thus,

$$\begin{aligned}\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] &= \frac{n_0}{n(n-1)n_1} \left\{ \sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \right\} \\ &= \frac{n_0}{n(n-1)n_1} \left\{ \sum_{i=1}^n X_i^2 - \frac{1}{n} (n\bar{X})^2 \right\} \\ &= \frac{n_0}{n(n-1)n_1} \left\{ \sum_{i=1}^n X_i^2 - n\bar{X}^2 \right\}.\end{aligned}$$

Variance Proof (SATE) - Answer 3 (5)

- To express this as a sum of centered squares, use

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2 \\ &= \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 \\ &= \sum_{i=1}^n X_i^2 - n\bar{X}^2.\end{aligned}$$

- Substituting this identity gives the desired result:

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{n_0}{n(n-1)n_1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Variance Proof (SATE) - Question 4

- Recall the potential-outcome expression for X_i :

$$X_i = Y_i(1) + \frac{n_1}{n_0} Y_i(0).$$

- Substitute this expression into the result of Question 3 to show

$$\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right).$$

Variance Proof (SATE) - Answer 4 (1)

- First express $\bar{X}$ in terms of the potential outcomes:

$$\begin{aligned}\bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i \\ &= \frac{1}{n} \sum_{i=1}^n Y_i(1) + \frac{n_1}{n_0} \cdot \frac{1}{n} \sum_{i=1}^n Y_i(0) \\ &= \bar{Y}(1) + \frac{n_1}{n_0} \bar{Y}(0).\end{aligned}$$

- Therefore,

$$\begin{aligned}X_i - \bar{X} &= \{Y_i(1) - \bar{Y}(1)\} \\ &\quad + \frac{n_1}{n_0} \{Y_i(0) - \bar{Y}(0)\}.\end{aligned}$$

Variance Proof (SATE) - Answer 4 (2)

- Substitute both potential-outcome deviations into the result of Question 3:

$$\begin{aligned}\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] &= \frac{1}{n} \left[\frac{n_0}{(n-1)n_1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] \\ &= \frac{1}{n} \left[\frac{n_0}{(n-1)n_1} \sum_{i=1}^n \left\{ (Y_i(1) - \bar{Y}(1)) \right. \right. \\ &\quad \left. \left. + \frac{n_1}{n_0} (Y_i(0) - \bar{Y}(0)) \right\}^2 \right].\end{aligned}$$

Variance Proof (SATE) - Answer 4 (3)

- Expand the square, keeping the cross-product term:

$$\begin{aligned} & \mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] \\ &= \frac{1}{n} \left[\frac{n_0}{(n-1)n_1} \sum_{i=1}^n \left\{ (Y_i(1) - \bar{Y}(1))^2 \right. \right. \\ & \quad \left. \left. + \frac{n_1^2}{n_0^2} (Y_i(0) - \bar{Y}(0))^2 \right. \right. \\ & \quad \left. \left. + 2 \frac{n_1}{n_0} (Y_i(1) - \bar{Y}(1))(Y_i(0) - \bar{Y}(0)) \right\} \right]. \end{aligned}$$

Variance Proof (SATE) - Answer 4 (4)

- Distribute the coefficient and recognize the finite-population variances and covariance:

$$\begin{aligned} & \mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] \\ &= \frac{1}{n} \left[\frac{n_0}{n_1} \cdot \frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \bar{Y}(1))^2 \right. \\ & \quad + \frac{n_1}{n_0} \cdot \frac{1}{n-1} \sum_{i=1}^n (Y_i(0) - \bar{Y}(0))^2 \\ & \quad \left. + \frac{2}{n-1} \sum_{i=1}^n (Y_i(0) - \bar{Y}(0))(Y_i(1) - \bar{Y}(1)) \right] \\ &= \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right). \end{aligned}$$

Variance Proof (SATE) - An Upper Bound

- Define the finite-population variance of treatment effects:

$$S_{\tau}^2 = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - Y_i(0) - \tau_{SATE}\}^2 = S_1^2 + S_0^2 - 2S_{01}.$$

- Substitute $2S_{01} = S_1^2 + S_0^2 - S_{\tau}^2$:

$$\begin{aligned}\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] &= \frac{1}{n} \left\{ \left(\frac{n_0}{n_1} + 1 \right) S_1^2 + \left(\frac{n_1}{n_0} + 1 \right) S_0^2 - S_{\tau}^2 \right\} \\ &= \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_{\tau}^2}{n}.\end{aligned}$$

- We cannot observe both potential outcomes for a unit, so S_{τ}^2 is not identified. Since $S_{\tau}^2 \geq 0$, dropping it gives an upper bound on the variance.

Variance Proof (PATE) - Question 5

- Assume potential-outcome pairs are sampled i.i.d. from a super-population, and use the result

$$\mathbb{E}[S_t^2] = \sigma_t^2, \quad t = 0, 1.$$

- Finally, derive the variance for PATE:

$$\mathbb{V}[\hat{\tau}] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

Variance Proof (PATE) - Answer 5 (1)

- Apply the law of total variance:

$$\begin{aligned}\mathbb{V}[\hat{\tau}] &= \mathbb{E}[\mathbb{V}[\hat{\tau} \mid \mathcal{O}_n]] + \mathbb{V}[\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n]] \\ &= \mathbb{E}\left[\frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + 2S_{01}\right)\right] + \mathbb{V}[\tau_{SATE}].\end{aligned}$$

- The first term is the average assignment variance; the second is the sampling variance of SATE, since $\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{SATE}$.

Variance Proof (PATE) - Answer 5 (2)

- Define the super-population variances and covariance:

$$\begin{aligned}\sigma_t^2 &= \mathbb{V}[Y_i(t)], \quad t = 0, 1, \\ \sigma_{01} &= \text{Cov}[Y_i(1), Y_i(0)].\end{aligned}$$

- With i.i.d. sampling, sample variances and covariance are unbiased:
 $\mathbb{E}[S_t^2] = \sigma_t^2$ and $\mathbb{E}[S_{01}] = \sigma_{01}$. The first term becomes

$$\mathbb{E}\left[\frac{1}{n}\left(\frac{n_0}{n_1}S_1^2 + \frac{n_1}{n_0}S_0^2 + 2S_{01}\right)\right] = \frac{1}{n}\left(\frac{n_0}{n_1}\sigma_1^2 + \frac{n_1}{n_0}\sigma_0^2 + 2\sigma_{01}\right).$$

Variance Proof (PATE) - Answer 5 (3)

- The second term is the variance of the sample average treatment effect:

$$\begin{aligned}\mathbb{V}[\tau_{SATE}] &= \mathbb{V}\left[\frac{1}{n} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}\right] \\&= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}[Y_i(1) - Y_i(0)] \\&= \frac{1}{n^2} \cdot n(\mathbb{V}[Y_i(1)] + \mathbb{V}[Y_i(0)] - 2 \text{Cov}[Y_i(1), Y_i(0)]) \\&= \frac{1}{n}(\sigma_1^2 + \sigma_0^2 - 2\sigma_{01}).\end{aligned}$$

- I.i.d. draws from the super-population imply zero covariance between units and the same variance for every unit.

Variance Proof (PATE) - Answer 5 (4)

- Combine both terms, writing out all six components:

$$\begin{aligned}\mathbb{V}[\hat{\tau}] &= \frac{n_0}{nn_1}\sigma_1^2 + \frac{n_1}{nn_0}\sigma_0^2 + \frac{2}{n}\sigma_{01} \\ &\quad + \frac{1}{n}\sigma_1^2 + \frac{1}{n}\sigma_0^2 - \frac{2}{n}\sigma_{01} \\ &= \left(\frac{n_0}{nn_1} + \frac{1}{n}\right)\sigma_1^2 + \left(\frac{n_1}{nn_0} + \frac{1}{n}\right)\sigma_0^2 \\ &= \frac{n_0 + n_1}{nn_1}\sigma_1^2 + \frac{n_1 + n_0}{nn_0}\sigma_0^2 \\ &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.\end{aligned}$$

- The $2\sigma_{01}/n$ terms cancel; use $n = n_0 + n_1$.

Neyman Variance Estimator

- Estimate each variance using observed outcomes ($n_t \geq 2$):

$$\bar{Y}_t^{\text{obs}} = \frac{1}{n_t} \sum_{i: T_i=t} Y_i, \quad \hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i: T_i=t} (Y_i - \bar{Y}_t^{\text{obs}})^2.$$

- Each arm is a random sample of the fixed units, so $\mathbb{E}[\hat{\sigma}_t^2 \mid \mathcal{O}_n] = S_t^2$.
The **Neyman variance estimator** is

$$\widehat{\mathbb{V}[\hat{\tau}]} = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}.$$

- Its conditional expectation is

$$\mathbb{E}[\widehat{\mathbb{V}[\hat{\tau}]} \mid \mathcal{O}_n] = \mathbb{V}[\hat{\tau} \mid \mathcal{O}_n] + \frac{S_\tau^2}{n}.$$

Conservative for SATE (unbiased if effects are constant); unbiased for PATE since $\mathbb{E}[\hat{\sigma}_t^2] = \sigma_t^2$.

Blocking: Setting

- We next consider whether blocking can reduce the variance of our estimator for **PATE**.
- There are J pre-defined blocks in an infinite population. Let B_i denote the block of unit i , and let

$$w_j = \mathbb{P}(B_i = j), \quad \sum_{j=1}^J w_j = 1.$$

- **Stratified random sampling:** sample $n_j = w_j n$ units independently from each block.
 - Within each block, potential-outcome pairs are sampled i.i.d.
- Use complete randomization independently within each block, with the **same treatment proportion:**

$$k = \frac{n_1}{n} = \frac{n_{1j}}{n_j}, \quad 0 < k < 1.$$

Blocking: Estimator

- First, calculate the difference-in-means estimator within each block:

$$\hat{\tau}_j = \frac{1}{n_{1j}} \sum_{i: B_i=j} T_i Y_i - \frac{1}{n_{0j}} \sum_{i: B_i=j} (1 - T_i) Y_i.$$

- Take the weighted average across blocks:

$$\hat{\tau}_{\text{block}} = \sum_{j=1}^J w_j \hat{\tau}_j.$$

- Each $\hat{\tau}_j$ is unbiased for the PATE within block j . Thus,

$$\begin{aligned} \mathbb{E}_{\text{block}}[\hat{\tau}_{\text{block}}] &= \sum_{j=1}^J w_j \mathbb{E}[Y_i(1) - Y_i(0) \mid B_i = j] \\ &= \mathbb{E}[Y_i(1) - Y_i(0)] = \tau_{\text{PATE}}. \end{aligned}$$

Variance Proof (Without Blocking)

- **Without blocking:** sample n units i.i.d. from the population and use complete randomization.
 - The numbers sampled from each block are now random.
 - Keep the same total sample size n and treatment proportion k .
- Recall the variance for PATE, where $\sigma_t^2 = \mathbb{V}[Y_i(t)]$:

$$\begin{aligned}\mathbb{V}[\hat{\tau}] &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0} \\ &= \frac{\sigma_1^2}{kn} + \frac{\sigma_0^2}{(1-k)n} \\ &= \frac{1}{n} \left(\frac{\sigma_1^2}{k} + \frac{\sigma_0^2}{1-k} \right).\end{aligned}$$

- We average over both **sampling** and **treatment assignment**.

Variance Proof (Blocking) - (1)

- Define the potential-outcome variance within block j :

$$\sigma_{tj}^2 = \mathbb{V}[Y_i(t) \mid B_i = j], \quad t = 0, 1.$$

- Apply the same PATE variance formula within each block:

$$\mathbb{V}_{\text{block}}[\hat{\tau}_j] = \frac{\sigma_{1j}^2}{n_{1j}} + \frac{\sigma_{0j}^2}{n_{0j}}.$$

- Substitute $n_{1j} = kn_j$ and $n_{0j} = (1 - k)n_j$:

$$\begin{aligned} \mathbb{V}_{\text{block}}[\hat{\tau}_j] &= \frac{\sigma_{1j}^2}{kn_j} + \frac{\sigma_{0j}^2}{(1 - k)n_j} \\ &= \frac{1}{n_j} \left(\frac{\sigma_{1j}^2}{k} + \frac{\sigma_{0j}^2}{1 - k} \right). \end{aligned}$$

Variance Proof (Blocking) - (2)

- Sampling and treatment assignment are independent across blocks. Therefore, the block estimators are **independent**:

$$\begin{aligned}\mathbb{V}_{\text{block}}[\hat{\tau}_{\text{block}}] &= \mathbb{V}_{\text{block}} \left[\sum_{j=1}^J w_j \hat{\tau}_j \right] \\ &= \sum_{j=1}^J w_j^2 \mathbb{V}_{\text{block}}[\hat{\tau}_j].\end{aligned}$$

- Substitute the within-block variance and use $n_j = w_j n$:

$$\begin{aligned}\mathbb{V}_{\text{block}}[\hat{\tau}_{\text{block}}] &= \sum_{j=1}^J \frac{w_j^2}{w_j n} \left(\frac{\sigma_{1j}^2}{k} + \frac{\sigma_{0j}^2}{1-k} \right) \\ &= \frac{1}{n} \sum_{j=1}^J w_j \left(\frac{\sigma_{1j}^2}{k} + \frac{\sigma_{0j}^2}{1-k} \right).\end{aligned}$$

Efficiency Gain due to Blocking - (1)

- Recall the **law of total variance**:

$$\mathbb{V}[X] = \mathbb{E}[\mathbb{V}[X \mid B]] + \mathbb{V}[\mathbb{E}[X \mid B]].$$

- Apply this to each potential outcome, with $t = 0, 1$:

$$\sigma_t^2 = \underbrace{\mathbb{E}[\mathbb{V}[Y_i(t) \mid B_i]]}_{\text{within-block variance}} + \underbrace{\mathbb{V}[\mathbb{E}[Y_i(t) \mid B_i]]}_{\text{across-block variance}}.$$

- The first term is a weighted average across blocks:

$$\mathbb{E}[\mathbb{V}[Y_i(t) \mid B_i]] = \sum_{j=1}^J w_j \sigma_{tj}^2.$$

- Thus, total variance is at least as large as within-block variance:

$$\sigma_t^2 - \sum_{j=1}^J w_j \sigma_{tj}^2 = \mathbb{V}[\mathbb{E}[Y_i(t) \mid B_i]] \geq 0.$$

Efficiency Gain due to Blocking - (2)

- Subtract the variance under the blocked design:

$$\begin{aligned} & \mathbb{V}[\hat{\tau}] - \mathbb{V}_{\text{block}}[\hat{\tau}_{\text{block}}] \\ &= \frac{1}{n} \left\{ \frac{\sigma_1^2}{k} + \frac{\sigma_0^2}{1-k} - \sum_{j=1}^J w_j \left(\frac{\sigma_{1j}^2}{k} + \frac{\sigma_{0j}^2}{1-k} \right) \right\}. \end{aligned}$$

- Collect the terms for $Y_i(1)$ and $Y_i(0)$ separately:

$$\begin{aligned} & \mathbb{V}[\hat{\tau}] - \mathbb{V}_{\text{block}}[\hat{\tau}_{\text{block}}] \\ &= \frac{1}{n} \left\{ \frac{\sigma_1^2 - \sum_{j=1}^J w_j \sigma_{1j}^2}{k} + \frac{\sigma_0^2 - \sum_{j=1}^J w_j \sigma_{0j}^2}{1-k} \right\}. \end{aligned}$$

Efficiency Gain due to Blocking - (3)

- Substitute the law-of-total-variance result:

$$\begin{aligned} & \mathbb{V}[\hat{\tau}] - \mathbb{V}_{\text{block}}[\hat{\tau}_{\text{block}}] \\ &= \frac{1}{n} \left\{ \frac{\mathbb{V}[\mathbb{E}[Y_i(1) \mid B_i]]}{k} + \frac{\mathbb{V}[\mathbb{E}[Y_i(0) \mid B_i]]}{1-k} \right\} \\ &\geq 0. \end{aligned}$$

- Both terms are nonnegative because they are variances, and $0 < k < 1$.
- $\rightarrow$ **Blocking weakly reduces the variance** in this setting.
 - The gain comes from differences in potential-outcome means across blocks.

Efficiency Gain due to Blocking: Intuition

- **What does blocking remove?**

- Without blocking, block composition varies in both treatment arms.
- This design fixes each block's share at w_j in both arms.
- Only the within-block component remains in the variance formula.

- **When is the gain larger?**

- When the potential-outcome means differ more across blocks.
- No gain if $\mathbb{E}[Y_i(t) \mid B_i = j]$ is the same across all blocks, for both $t = 0$ and $t = 1$.

- **What are we comparing?**

- Two sampling-and-assignment designs, not two analyses of the same blocked experiment.
- With a common k , the overall difference in means in the blocked experiment already equals $\sum_j w_j \hat{\tau}_j$.

Matched-Pair Design: Review

- **Motivation**

- Blocking improve efficiency → Can we keep blocking?

- **Procedure**

- Create $J = n/2$ similar units
- Randomize treatment assignment within each pair

- **Estimator**

$$\hat{\tau}_{\text{pair}} = \frac{1}{J} \sum_{j=1}^J W_j (Y_{1j} - Y_{2j})$$

where

- Y_{1j} is the first outcome in j -th pair
- Y_{2j} is the second outcome in j -th pair
- $W_j = 1$ if first unit received treatment and $W_j = -1$ if second unit received treatment

Matched-Pair Design: Variance

- **Variance**

$$\mathbb{V}(\hat{\tau}_{\text{pair}}) = \underbrace{\frac{\sigma_1^2}{J} + \frac{\sigma_2^2}{J}}_{\text{Var w/o design}} - \underbrace{\frac{2}{J}\text{Cov}(Y_{1j}, Y_{2j})}_{\text{within-pair covariance}}$$

where “Var w/o design” is the variance under complete randomization.

- In other words,

$$\mathbb{V}(\hat{\tau}) = \frac{\sigma_1^2}{J} + \frac{\sigma_2^2}{J} = \mathbb{V}(\hat{\tau}_{\text{pair}}) + \frac{2}{J}\text{Cov}(Y_{1j}, Y_{2j})$$

- Notice that $n_1 = n_0 = J$ (within each pair, one treated and one control)

Matched-Pair Design: Variance Estimator

- **Variance Estimator**

$$\widehat{\mathbb{V}}(\hat{\tau}_{\text{pair}}) = \frac{1}{J(J-1)} \sum_{j=1}^J \left(W_j(Y_{1j} - Y_{2j}) - \hat{\tau}_{\text{pair}} \right)^2$$

- **Question:** How to optimize matching so that we can maximize efficiency?
 - There are so many ways to pair observations
 - Higher within-pair covariance leads to more efficiency gains
 - We want to create pair in which all pairs are similar enough

Optimal Matching: Idea

- First, calculate the distance between any two units
 - We use **Mahalanobis distance**, in which distance between observation i and j is defined as

$$D(X_i, X_j) = \sqrt{(X_i - X_j)^T \text{Var}[X]^{-1}(X_i - X_j)}$$

- Intuition: Distance between covariates normalized by their variance
- **Goal:** Find the matching by which we can minimize the sum of distance

Greedy v.s. Optimal Matching

- Let's think with an example.
- Suppose there are 4 people in the world.

	1	2	3	4
1		1	9	5
2	1		5	9
3	9	5		12
4	5	9	12	

- NOTE: Diagonal element is missing to avoid the matching of the same unit.

Greedy Matching: 2 (a)

- Consider the greedy matching in 2 (a)
 - First, compute the distance between all pairs of health clusters.
 - Based on this distance matrix, select two clusters which are most similar and set them aside as a match

	1	2	3	4
1		1	9	5
2	1		5	9
3	9	5		12
4	5	9	12	

- You first select 1 (matching between unit 1 and 2)

Greedy Matching: 2 (a)

- After matching unit 1 and 2, you cannot use unit 1 or 2 for the future matching
 - For example, matching unit 1 and 3 is infeasible since unit 1 is already matched with unit 2

	1	2	3	4
1		1	9	5
2	1		5	9
3	9	5		12
4	5	9	12	

- In this case, you need to match unit 3 and 4.
- Sum of total distance: $1 + 12 = 13$

Greedy Matching: 2 (b)

- Consider the greedy matching in 2 (b)
 - Randomly select a cluster and then find the cluster which is most similar to it.
 - Set them aside as a matched pair.
- Suppose that you randomly pick up unit 2.
 - The closest for unit 2 is unit 1.
- As you match unit 2 and 1, you cannot use these units for the future matching.

	1	2	3	4
1		1	9	5
2	1		5	9
3	9	5		12
4	5	9	12	

- Sum of total distance: $1 + 12 = 13$

Optimal Matching

- Greedy matching is suboptimal
 - In the previous table, minimum distance is 10 ((1, 4) and (2, 3))
- **Optimal Matching:** Directly minimize the sum of distance
 - Algorithm: Optimal nonbipartite matching
- Optimization problem is written as

$$\begin{aligned} \min_M \quad & \sum_{i=1}^n M_{ij} D_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^n M_{ij} = 1, \quad \sum_{j=1}^n M_{ij} = 1 \end{aligned}$$

- Constraint: Each unit is used for matching only once

	1	2	3	4
1		1	9	5
2	1		5	9
3	9	5		12
4	5	9	12	